# E.F. Johnson 153-210 (Type D) 3-Gang Variable Capacitor

### E.F. Johnson Company / Cardwell Johnson Capacitor Division - 153 Series

## Product Overview

The E.F. Johnson 153-210 is a heavy-duty, commercial-grade Type D (153 series) 3-gang air variable capacitor. Designed for multi-stage RF amplifiers, multi-band receiver front-ends, and transmitter stages, it features 0.051 inch thick aluminum plates with rounded edges to prevent corona discharge, supported on massive 5/16 inch aluminum tie rods and steatite insulators.

## Technical Specifications

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| **Part Number** | 153-210 (Stamped 153210) |
| **Series** | Type D - 153 Series Variable Air Capacitors |
| **Gangs / Sections** | Triple-Gang (3-Section ganged tuning) |
| **Plates Specification** | Heavy 0.051 inch aluminum plates with rounded edges for high breakdown voltage |
| **Tie Rods** | 5/16 inch diameter aluminum tie rods for rigid frame assembly |
| **Shaft** | 1/4 inch cadmium-plated steel shaft, extended for drive coupling |
| **Insulation** | High-frequency steatite ceramic stator insulation strips |
| **Contact Springs** | Laminated phosphor bronze wiper springs for smooth rotor grounding |

## Key Features & Construction

- Three matched sections on a common rotor shaft for synchronized stage tuning
- Thick 0.051 inch aluminum plates with smoothed edges for high breakdown voltage
- Rugged 5/16 inch aluminum tie-rod frame preventing mechanical deflection and microphonics
- Rear bearing adjustment screw with locknut for zero axial shaft play
- Independent stator terminal solder lugs for each of the three gangs

## Mechanical & Mounting Details

- Frame dimensions: Width 4-1/4 inches, Height 4 inches
- Shaft center height: 2-5/32 inches from mounting base
- Mounting: Front/rear end frames tapped 8-32 (three holes) for panel or bracket mounting
- Rotor shaft: 1/4 inch diameter extended shaft suitable for vernier dials or flexible couplings